

NEERAJ KUMAR

+91-8800119639 reachoutneerajkumar@gmail.com [Portfolio](#) [LinkedIn](#) [GitHub](#) [LeetCode](#)

EDUCATION

Guru Gobind Singh Indraprastha University, New Delhi
Bachelor's Degree - Automation & Robotics - CGPA - 7.00

Nov 2022 – Jul 2026
New Delhi

TECHNICAL SKILLS

Languages: C++17/20, C, Python, JavaScript

Technologies/Frameworks: React, Node.js, Express, PostgreSQL, MongoDB, Redis, WebSockets, Firebase, Docker, Linux/UNIX environments, REST APIs, TCP/IP

Tools: Git, Nginx, GitHub Actions, Vercel, Postman, VS Code

Core Competencies: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, System Design, Distributed Systems

EXPERIENCE

Outlier - Code Reviewer & AI Data Annotator

Mar 2026 – Apr 2026

Freelance

Remote

- Reviewed and evaluated AI-generated code and technical responses for correctness, logic, edge cases, and adherence to task requirements.
- Annotated and ranked model outputs, identifying logical errors, inefficient implementations, and inconsistencies to improve code-generation quality.
- Applied structured evaluation criteria across programming and technical tasks, providing detailed feedback and corrected solutions.

PROJECTS

FixEngine | [GitHub](#) | C++20, CMake, Linux

May 2026 – Sep 2026

- Built a zero-allocation **FIX protocol parsing engine** in C++20 from scratch, implementing a 5-stage pipeline spanning ingestion, tokenization, parsing, validation, and orchestration.
- Designed a stateful incremental ingestion buffer for arbitrarily chunked FIX messages, returning explicit **MessageReady**, **NeedMoreData**, **BufferFull**, and **InvalidMessage** states instead of exception-based control flow.
- Implemented **checksum validation and BodyLength verification** with explicit error types, maintaining a deterministic and allocation-free critical processing path.

RedisCLI | [GitHub](#) | C++17, POSIX Sockets, RESP2

Jan 2026 – May 2026

- Built a Redis CLI client from scratch in C++17, implementing the full **RESP2** protocol serializer and parser.
- Implemented event-driven Pub/Sub mode using `poll()` to multiplex stdin and Redis socket simultaneously - streaming published messages while keeping terminal input responsive via GNU Readline callback API.

Options Pricing Simulator | [GitHub](#) | Python, SciPy, NumPy

Jan 2026 – Apr 2026

- Developed a **Monte Carlo** simulation engine to price **European Options** with configurable simulation runs.
- Validated results against Black-Scholes analytical model, achieving **<0.03** pricing deviation at **10,000** simulations.
- Computed option sensitivities : Delta (Δ), Vega (ν), Gamma (Γ), and Theta (Θ) - to support quantitative risk analysis.

Discord-Vault | [GitHub](#) | Python, SHA256

Jan 2026 – Apr 2026

- Developed a secure Discord-based file vault for storing and retrieving encrypted files through a bot interface.
- Implemented SHA-256 based integrity verification to detect file tampering and ensure data consistency.
- Implemented lossless PNG compression and decompression, reducing image storage overhead while preserving pixel-perfect data integrity.
- Automated file upload, retrieval, and vault management through Discord commands, eliminating the need for a dedicated frontend.

CERTIFICATIONS

- Postman API Fundamentals Student Expert** - [Postman](#)
- Career Essentials in GitHub Professional Certificate** - [LinkedIn Learning](#)
- Software Engineering Job Simulation (JPMorgan Chase & Co.)** - [Forge](#)

ACHIEVEMENTS & EXTRACURRICULARS

- Achieved Max Rating of **1614** on [LeetCode](#); solved **300+** algorithmic problems on LeetCode in C++.
- Achieved Global Rating of **123** on CodeChefs and max rating of **1159** on [Codeforces](#).
- Represented the college Volleyball team in inter-collegiate tournaments and secured **3rd place**.